

Global Facilities - Design & Construction Standard - Mechanical – HPM Chemical / Gas Room HVAC System

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1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new HPM Chemical / Gas Room HVAC Systems as part of new Greenfield FAB Construction projects.
- 1.2 The standard is applicable for HVAC Systems provided in HPM Chemical / Gas Rooms.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 This document covers the new Basebuild System installations for buildings only, as part of new Greenfield FAB Construction projects. This document does NOT cover Tool Installation or Point-of-Use (POU) Abatement.
- 2.2 This document does not cover other General HVAC or Exhaust systems, such as General Building Areas HVAC, FAB Process Exhaust or General Building Areas Exhaust, Smoke Exhaust, or other Specialized Exhaust systems.
- 2.3 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- **ACH** - Air Changes per Hour
- **ACMV** - Air Conditioning and Mechanical Ventilation
- **AE** - Acid Exhaust
- **AHJ** - Authority having Jurisdiction
- **AME** - Ammonia Exhaust (also known as Alkali Exhaust or Caustic Exhaust)
- **CDU** - Chemical Distribution Unit
- **DOL** - Direct On-Line
- **EDPM** - Ethylene Propylene Diene Monomer
- **EXH** - Exhaust
- **FMCS/SCADA** - Facility Monitoring and Control System / Supervisory Control and Data Acquisition
- **FRP** - Fiber Reinforced Plastic
- **FCT** - Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
- **GFTT** - Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC** - Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team** - This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** - This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team** - This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team** - This is used interchangeably with Global Facilities Line Planning Team
- **GC** - Gas Cabinet
- **GE** - General Exhaust
- **GHVAC** - General Heating, Ventilation, Air Conditioning
- **HAP** - Hazardous Air Pollutants
- **HPM** - Hazardous Production Material
- **MAU** - Make-up Air Unit
- **MUA** - Make-up Air
- **ICW** - Industrial City Water
- **LCP** - Local Control Panel
- **LOTO/COHE** - Lockout Tagout / Control of Hazardous Energy
- **N+1+F** - System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, + future unit(s)

- **PLC** - Programmable Logic Controller
- **POC/LPOC** - Point of Connection / Lateral Point of Connection
- **POU** - Point of Use
- **SMACNA**
- **SE** - Solvent Exhaust (also known as VOC Exhaust or Organic Exhaust)
- **TUM** - Tool Utility Matrix
- **TMA** - Trimethylaluminium
- **UPS** - Uninterruptible Power Supply
- **UPW/DI/RO** - Ultrapure Water / De-Ionized Water / Reverse Osmosis Water
- **VMB** - Valve Manifold Box
- **VOC** - Volatile Organic Compounds
- **VSD/VFD** - Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

The following Exhaust and Make-up air (MAU) systems are used to support the requirements from Manufacturing and Facilities loads:

- 5.1 **Chemical Rooms** are buildings or the portion of it, or exterior area used for the storage of hazardous materials in excess of exempt amounts.
- 5.2 **Hazardous Production Material (HPM)** is a solid, liquid, or gas associated with semiconductor manufacturing that has a degree-of-hazard rating of 3 or 4 in health, flammability, instability, or water reactivity in accordance with NFPA 704 and that is used directly in research, laboratory, or production processes that have as their end product materials that are not hazardous.
- 5.3 **Acid exhaust (AE)** is used to remove and neutralize acidic compounds from tool process exhaust via abatement scrubbers, before discharging the exhaust to atmosphere as clean air.

The hazardous air pollutants (HAP) that are abated by the Acid Exhaust system include Chlorine (Cl), Hydrochloric acid (HCl), Hydrofluoric acid (HF), Phosphoric acid (H₃PO₄), Hydrobromic acid (HBr), Nitric acid (HNO₃), Sulfuric acid (H₂SO₄), and Acetic acid (CH₃COOH).

- 5.4 **Ammonia exhaust (AME)** is used to remove and neutralize ammonia (NH₃, NH₄) compounds from tool process exhaust via abatement scrubbers, before discharging the exhaust to atmosphere as clean air. Ammonia Exhaust is also referred to as Alkali Exhaust or Caustic Exhaust.
- 5.5 **Solvent exhaust (SE)** is used to remove and destroy Volatile Organic Compounds (VOC) from tool process exhaust via VOC abatement units, before discharging the exhaust to atmosphere via clean air. The typical VOCs include Isopropyl Alcohol (IPA, C₃H₈O), Methyl Propyl Acetate (PGMEA, C₆H₁₂O₂), Butane (C₄H₁₀), and Benzene (C₆H₆). Solvent Exhaust is also referred to as VOC Exhaust or Organic Exhaust.
- 5.6 **General Exhaust (GE)** is used to remove non-corrosive, non-toxic, non-hazardous, or heated exhaust from production tools process exhaust and then discharging the exhaust directly to atmosphere without any processing or abatement.

The typical uses for General Exhaust include heat exhaust, cabinet enclosure exhaust, and chamber exhaust from tools including Diffusion, FAB Support equipment.

- 5.7 **Makeup Air (MUA)** is 100% outside air that is supplied to provide control of indoor air temperature & humidity, room pressurization level and to provide fresh air for occupants. It compensates for the exhausted air via various exhaust systems or natural exfiltration. The air is conditioned, filtered, and supplied via the Makeup Air Units (MAU).
- 5.8 **General HVAC (GHVAC) Exhaust** is used to remove non-corrosive, non-toxic, non-hazardous, or smoke-logged indoor air from the room / area / zone to facilitate air changes rate. This system is a critical part of the room's air conditioning and mechanical ventilation (ACMV) system. Not to be confused with General Exhaust (GE) which primarily serves as Production Tools process exhaust.

6.0 Performance Specification and System Provision

- 6.1 The HPM Chemical / Gas Room HVAC Systems shall be designed and installed to achieve the required minimum room ventilation rate, room condition and the ACMV & exhaust system arrangement as outlined below; See **Table 1, Table 2 and Table 3**.
- 6.2 The Normal mode exhaust (ventilation) is continuous exhaust used during normal operation. This consists of exhaust from HPM equipment such as Chemical Tanks, Chemical Distribution Units (CDU), Gas Cabinets, VMBs, and normal room exhaust.
- 6.3 The Emergency mode exhaust is additional exhaust that is activated during emergency scenarios (i.e. Chemical or Gas leak, fire event, etc.), which provides higher air change rates to purge the space from harmful, corrosive, or toxic fumes.
- 6.4 Emergency mode exhaust shall be activated in the following ways:-
- 6.4.1 Automatically in a fire event or Chemical / Gas leak event.
 - 6.4.2 Remotely by Operation through SCADA.
 - 6.4.3 Locally through a manual emergency switch (e.g. break-glass) triggered by on-site Operator.

Table 1: Minimum Room Ventilation Rate and Exhaust Systems Arrangement

HPM Room	Room Ventilation Rate, (Air Changes per Hour (ACH)) *Note 2		Exhaust System	
	Normal Mode	Emergency Mode	Central Exhaust System (for Normal Exhaust, unless otherwise noted)	Dedicated Exhaust System (for Emergency Exhaust, unless otherwise noted) *Note 4
<u>Chemical</u>				
Caustic (Bases) / Caustic Dispense rooms	6	20	<u>Normal Exh:</u> AME	<u>Emergency Exh:</u> GHVAC
Acids / Acid Dispense rooms	6	20	<u>Normal Exh:</u> AE	<u>Emergency Exh:</u> GHVAC
Chemical Storage Rooms	6	20	<u>Normal Exh:</u> AE, AME, SE	<u>Emergency Exh:</u> GHVAC
Solvent / Solvent Dispense rooms	6	20	<u>Normal Exh:</u> SE	<u>Emergency Exh:</u> GHVAC
Oxidizer rooms	6	20	<u>Normal Exh:</u> AE	<u>Emergency Exh:</u> GHVAC
Slurry Storage Rooms	6	20	<u>Normal Exh:</u> AE, AME	<u>Emergency Exh:</u> GHVAC
<u>Gas</u>				
Inert Gas Rooms	6	20	<u>Normal Exh:</u> GE	<u>Emergency Exh:</u> GHVAC
Corrosive Gas Rooms	6	20	<u>Normal Exh:</u> AE	<u>Emergency Exh:</u> GHVAC
Flammable Gas Rooms	6*	20	*Note 3	<u>Normal/Emergency:</u> Required, Duty-Standby
CIF3/HF Gas Rooms	6	20	<u>Normal Exh:</u> AE	<u>Emergency Exh:</u> GHVAC
TMA Rooms	6	20	<u>Normal Exh:</u> AE	<u>Emergency Exh:</u> GHVAC
Specialty Gas Rooms	6	20	<u>Normal Exh:</u> AE	<u>Emergency Exh:</u> GHVAC

- Note 1:** The Room Ventilation and Exhaust System shall be designed and installed with capability to meet the required minimum air change stated above, or in compliance with the mandatory ventilation rate by Local AHJ, whichever is higher or more stringent. Dedicated & specialty exhaust system (e.g. CIF3 Exhaust) might also be mandatorily required by Local AHJ in certain country.
- Note 2:** Room Ventilation Rate is the total required ventilation rate in the room / zone. It is the summation of the ventilation rate through the Central Process Exhaust System and the General HVAC exhaust system. While General HVAC exhaust system can be shared among rooms through common header, the Dedicated Exhaust System serves only the rooms with similar hazard classification.
- Note 3:** Flammable Gas Room shall not be connected to any Central Exhaust System. For both the Normal and Emergency Room Exhaust, the Flammable Gas Room shall have its own dedicated Exhaust Fan(s)
- Note 4:** For some Local AHJ, such as Taiwan, that requires the emergency exhaust air to be abated (due to it might be polluted with HPM gases or vapors), the emergency exhaust shall be connected to the appropriate exhaust scrubbers or abatement units.

Table 2: Required Room Condition and ACMV System Arrangement

HPM Room	Room Condition (See Note*)		Remarks
	Room Temperature, °C	Relative Humidity, RH%	
<u>Chemical Rooms</u>			
Caustic (Bases) / Caustic Dispense room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Acids / Acid Dispense room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Chemical Storage Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Solvent / Solvent Dispense room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)

Oxidizer room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Slurry Storage Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
<u>Gas Rooms</u>			
Inert Gas Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Corrosive Gas Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Flammable Gas Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
CIF3/HF Gas Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
TMA Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)
Specialty Gas Room	20 ~ 25	≤65	• 100% Outdoor fresh air (single pass) • Negative Pressurized (-3 ~ -7 Pa)

Note*: The HVAC System shall be designed and installed with capability to meet the required room condition stated above, or in compliance with those required as recommended by Chemical or Gas Manufacturer, whichever is more stringent. The local climate profile shall also be taken into consideration to design an energy efficient system as recommended (or mandatory) by AHJ.

Actual site temperature and humidity specification for each room, and the actual operation of HVAC system for each room shall meet the requirement of the specific Chemicals or Gases that are stored / dispensed from the rooms, as specified by the Chemical or Gases Manufacturer.

Table 3: Special Requirement

Area / System	Description
TMA Room	Room temperature condition shall not be lower than 17°C to avoid the freezing of TMA

Note: The special requirement is specified as per the best knowledge of the common industrial practices. The actual design shall consider and apply this wherever / whichever (including but not limited to) is reasonably practicable.

6.5 **System Design** – Heating, Ventilating & Air Conditioning (HVAC) systems are required to achieve the performance specified above. While the detail design is on a project by project basis, the following concept should be captured:

- 6.5.1 Central Process Exhaust Systems should be used to exhaust the air from the HPM Chemical / Gas cabinet or enclosure. The air originated from these sources typically require air treatment (through scrubber or abatement) prior discharge to the atmosphere.
- 6.5.2 Dedicated Exhaust System should be used where there is a risk of spreading of hazard (e.g. fire risk due to the leaked flammable gas). This system is typically designed to exhaust the air to prevent accumulation of hazardous gas / smoke in the room (or zone).
- 6.5.3 Dedicated Exhaust System is allowed to serve more than a single room (or zone) if these areas are categorized to have similar hazard, and if such a design is permitted by AHJ and in compliance with relevant standard / code.
- 6.5.4 During emergency of a fire event or gas leak from a room, the Emergency (dedicated) exhaust fan shall energize and the motorized damper for that room shall open. The motorized dampers for the other rooms shall remain closed to isolate the rooms.
- 6.5.5 MUA supply shall be designed accordingly to facilitate the required ventilation rate and proper pressurization of the room air. MAU system is typically designed to provide cooling (through cooling coils) and heating (through heating coils or in-duct heater) to the room. MAU system is allowed to serve more than a single room (or zone).

- 6.5.6 While sharing of facilities as per 6.5.3 & 6.5.4, ductworks and MAU cooling & heating coils shall be designed with fittings (e.g. motorized dampers, sensing elements, control valves, etc) to have proper provision to ventilate, pressurize and regulate individual HPM Chemical / Gas Rooms' ACH and room air condition.
- 6.5.7 MAU system shall not be supplied from Fab Cleanroom MAU system. Provide MAU units separate from Fab MAU cleanroom system to allow for the independent air balancing control, and prevent any impact to Fab Cleanroom.
- 6.5.8 While sharing of facilities as per 6.5.3 & 6.5.4, fan motors should be designed with VSD to regulate its speed to operate at designated output corresponding to the required Normal / Emergency mode ACH for each rooms (or zones).
- 6.5.9 Exhaust system for HPM Chemical / Gas Rooms should be running in Normal Mode with Central Exhaust System first (if applicable), to provide the required Normal Mode Air Change Rate.
- 6.5.10 During Emergency Mode, the Emergency (Dedicated) Exhaust System will ramp up to the elevated rate, to provide the required Emergency Mode Air Change Rate. The room air change rate will then be increased together with Motorized Dampers actuation and MAU fans ramping up.

6.6 Performance Specification of MAU

- 6.6.1 While the design of MAU is highly dependent on the local weather profile, design parameters listed in **Table 4** serves as the guideline value to maintain the design consistency across Micron Fabs.

Table 4: Performance Specification for MAU

Parameter	Specification Requirement	Remarks
Supply Air Temperature	16 ~ 23°C (60.8 ~ 73.5 °F)	- Measured at outlet of MAU - Subject to local atmospheric condition
Supply Air Dewpoint	12 ~ 15°C (53.6 ~ 59 °F)	Measured at outlet of MAU (if humidity control is required)
External Static Pressure	500 - 800 Pa	- Measured at outlet of MAU - Subject to the ducting design

Note: The MAU System shall be designed and installed with capability to meet the required performance specification stated above. Actual operational parameters should be considered on a case-by-case basis as per the local atmospheric condition, room temperature and humidity requirements, and actual ducting design.

7.0 Equipment Sizing & Design Requirements:

7.1 **Redundancy** – Exhaust and MAU System main equipment shall be designed for N+1 redundancy.

- 7.1.1 For the dedicated exhaust system serving a single room (or single zone), the exhaust fans shall be in the arrangement of N+1 (1 duty and 1 standby) per the design capacity of that particular room (or single zone).
- 7.1.2 For the dedicated exhaust system serves a group of rooms (or zones) with similar hazard classification, the fans shall be in the arrangement of N+1 (1 duty and 1 standby) per the design capacity of the largest compartment of the rooms (or zones).
- 7.1.3 For the MAU system serves a group of rooms (or zones) with similar hazard classification, the units shall be in the arrangement of N+1 (1 duty and 1 standby) per the design capacity of the largest compartment of the rooms (or zones).
- 7.1.4 The design redundancy shall be in compliance with the requirement of AHJ if it is more stringent than 7.1.1, 7.1.2 & 7.1.3.

7.2 **System Diversity and Safety Factor** – Exhaust System sizing shall be selected so that the N capacity meets 115-120% of the exhaust demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM, plus the Facilities Basebuild requirements. Therefore, the Exhaust system sizing capacity for N capacity shall have 15-20% spare future capacity over the exhaust demand at full build (after initial construction of the system). Refer to the Global Facilities – Mechanical – Process Exhaust System Standard for additional requirements.

- 7.2.1 To determine the Exhaust demand at N full build, a diversity factor of 100% of the Peak demand from the TUM shall be used.
- 7.2.2 The Facilities Basebuild requirements shall include all other users of exhaust such as Gas Cabinets, Gas and Chemical VMBs, Chemical Supply Equipment, UPW and Wastewater Tanks and Equipment, Room Area exhaust, etc.
- 7.2.3 Make-up air shall be considered accordingly to compensate for the exhausted air, to maintain required room condition, as well as to regulate the room air pressure.
- 7.2.4 See **Tables 5** and **Table 6** for Exhaust system sizing criteria and calculation, while **Table 7** for MAU system sizing criteria and calculation.

Table 5: System Sizing Calculation – Acid, Alkali, General, & Solvent Exhaust

- Exhaust System Demand at Full Build, D:
 $D = (\text{TUM Peak Demand} \times 100\% \text{ Diversity}) + \text{Facilities Basebuild Requirements}^*$
- Exhaust System N capacity:
 $N = 120\% \times D$

Table 6: System Sizing Calculation – GHVAC Exhaust, for Emergency (Dedicated) Exhaust Sizing

With reference to Table 1

- Emergency (Dedicated) Exhaust System for Single Room Demand at Full Build, D:
 $D = \text{Minimum Emergency Mode Exhaust Rate} - \text{Total Normal Mode Exhaust Rate (via Central Process Exhaust system)}.$
- Emergency (Dedicated) Exhaust System for a Group of Rooms / Compartment Demand at Full Build, D:
 $D = \text{Minimum Emergency Mode Exhaust Rate} - \text{Total Normal Mode Exhaust Rate (via Central Process Exhaust System)}$
**For the room / compartment which gives the highest D
- Exhaust System N capacity:
 $N = 120\% \times D$

Table 7: System Sizing Calculation – MAU System

- MAU System Demand at Full Build, D:
 $D = \text{Total Normal Exhaust} + \text{Total Emergency Exhaust}$
 1. Total Normal Exhaust = Total Exhaust in Normal mode inclusive Central Exhaust System / Dedicated Exhaust System for ALL the rooms being served by the MAU.
 2. Total Emergency Exhaust = Total Exhaust in Emergency mode inclusive Central Exhaust System / Dedicated Exhaust System for the LARGEST (highest airflow) room / compartment being served by the MAU.

**Actual MAU output should be lower than the Actual Exhaust Rate at each rooms / compartment to maintain negative room pressure.
- MAU System N capacity:
 $N = 120\% \times D$

- 7.3 **Equipment Quantity** – Exhaust and MAU System main equipment shall be selected for a minimum of 2 units (1+1) or more, with each fan/unit to be equal in capacity sizing.
- 7.4 **Equipment Layout** – Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment, and to allow move-in and move-out of large components during maintenance and repair.
- 7.4.1 Exhaust Fans shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on fan assembly, impeller, and motor.
- 7.4.2 MAU shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on filters, coils, fan assembly, impeller, and motor.
- 7.4.3 Equipment should be connected to a common header / plenum in such a way that minimizes the pressure drop, and to optimize the redundancy of the entire system.
- 7.5 **Equipment Location** - The location of the fans shall be designed as such to reduce the length and pressure drop of the ductworks.
- 7.5.1 For the dedicated exhaust system serves single room (or zone), it is recommended to locate the exhaust fans at the common area not for occupants' circulation (e.g. rooftop mechanical area). The installation is planned for accessibility for inspection and maintenance activities on the fans. The ductworks should be planned to avoid passing other room (or zone) to minimize the risk of spreading of hazards (e.g. fire / smoke / leaked gas).
- 7.5.2 For the dedicated exhaust system serves a group of rooms (or zones) with similar hazard classification, it is recommended to locate the exhaust fans at the common area not for occupants' circulation (e.g. rooftop mechanical area). The common ductworks shall be routed in common area and should be planned to avoid passing other room (or zone) to minimize the risk of spreading of hazards (e.g. fire / smoke / leaked gas).
- 7.5.3 MAU is recommended to be located in an indoor equipment room with sufficient and accessible space for operational and maintenance activity. The MAU shall not be inside the HPM Chemical / Gas Room. The location of the MAU inlet louver for outdoor air should carefully be selected to avoid receiving dusty or hazardous outdoor air, or cross contamination from exhaust streams, during operation.

- 7.6 **Equipment Location, Exhaust Discharge Stacks** – Exhaust stacks shall be located away from Makeup Air Unit (MAU) intakes so that air discharged from the stacks is not re-entrained into the MAU intakes.
- 7.6.1 Exhaust Discharge Stack height shall comply with local/regional/national code, AHJ, and/or air emissions regulations requirements.
- 7.6.2 Computational Fluid Dynamics (CFD) Modeling shall be performed to simulate the path of exhaust discharge from the stacks. The modeling shall be used to ensure that exhaust discharge is not re-entrained into the MAU intakes.
- 7.7 **Equipment Arrangement, GHVAC Exhaust Fans** – GHVAC Exhaust Fans serving the same room/zone/multiple rooms shall be arranged in parallel together and connected via a common duct header (for N+1 configuration). This allows any combination of equipment to be in operation concurrently and allow for redundancy on all equipment, serving the same room/zone/multiple rooms.
- 7.7.1 Exhaust Fans shall be connected to a common ductwork header upstream of the fans. The header shall be constant diameter size across the entire length of the header.
- There shall multi-blade type motorized control damper and manual blastgate on the inlet ductwork to the fan. The control damper shall be programmed to open upon fan startup, and close during fan shutdown or fault.
 - The fans can be connected to a dedicated discharge stack or common discharge stack shared between the two fans.
 - It is recommended to keep a minimum of 3- to 5-duct diameters of unobstructed straight duct upon entrance into the exhaust fan inlet, to prevent any fan system effects or pre-swirling and turbulence.
- 7.8 **Pressure Drop Calculations** – Submit pressure drop calculations for each ductwork system, to show that each fan selection and the fan parameters (flow, pressure, power, etc.) are adequate to meet the design requirements.
- 7.9 **Equipment Selection Criteria** See **Tables 8** for Exhaust system and **Table 9** for MAU system selection criteria and requirements.
- 7.9.1 Exhaust fans dedicated to serve storage rooms that store flammable material (gas, or liquid forms), or potentially flammable (e.g. volatile flammable vapor), shall be equipped with protection device / installation that is sufficient and appropriate for (at least) Class 1, Division 2 hazardous area.

- 7.9.2 All electrical installations within the rooms shall be "spark proof", so that normal operation of the installation does not provide a source of ignition.
- 7.9.3 Subject to actual design criteria, higher level of hazard classification (and thus higher installation's protected type) should be considered wherever it is applicable.

Table 8: Exhaust Fan Equipment Selection Criteria – Acid, Ammonia, Solvent, General Exhaust, and GHVAC Exhaust

Equipment	Type & Description
Exhaust Fans – All (Acid, Ammonia, Solvent, and General Exhaust, GHVAC Exhaust)	• Exhaust Fans shall be Centrifugal type, Direct-Drive type. • Exhaust Fan sizing and selection shall be as follows: ○ Fans shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower). ○ Fan curve shall show the performance indicated at the maximum operating point within the fan motor size limitations (flow and pressure, based on fan impeller size, VSD speed, and full brake-horsepower to match motor size rating). Resultant fan tip speed shall be stated. ○ The maximum allowable fan impeller tip speed rating shall be published. ○ The fan maximum operating point tip speed shall be at least 15% - 20% less than the maximum allowable tip speed of the fan. Meaning, for continuous operation at the fan maximum operating point, the fan impeller tip speed shall be less than 80% - 85% of the maximum allowable fan impeller tip speed rating. ○ Fan impeller shall be AMCA rated, Class IV ○ Heavy duty bearings ○ Shaft shall have critical speed 125% of fans maximum operating speed ○ Mechanical efficiency shall be 75% above. ○ Fan assembly shall be statically and dynamically balanced. ○ Totally enclosed fan-cooled motor with over-temperature protection. ○ Sound level performance to be < 75 dBA at distance of 1m away from fan enclosure.

- Fan motor sizing and selection shall be as follows:
 - Fan motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions.
 - Fan motors shall be rated for inverter duty with variable speed operation.
 - Fan motor efficiency rating shall be minimum IE3.
 - Fan motor IP rating shall be minimum IP55. If located outdoors, fan motor IP rating shall be IP56.
- Fan installation shall be as follows:
 - Each Exhaust Fan shall have concrete inertia base and vibration isolation springs.
 - Each Exhaust Fan housing shall have access door to allow inspection and cleaning of fan impeller.
 - Each Exhaust Fan housing shall have drain connection port.
 - Each Exhaust Fan shall have drain piping connected from the drain connection port.
 - Each exhaust fan shall have minimum 25mm (1-inch) test ports upstream and downstream of the fan to allow airflow and pressure readings.
 - Each exhaust fan shall have minimum 150mm (6-inches) cleaning port upstream and downstream of the fan to allow inspection and cleaning.
 - Each Fan shall have flexible ductwork connection on inlet side and flexible ductwork connection on discharge side.

Acid and Ammonia Exhaust Fans

- Fan housing shall be all Fiberglass materials of construction.
- Fan impeller shall be made of Fiberglass, Carbon Fiber, or hybrid combination of Fiberglass and Carbon Fiber.

Note: Fan impellers made of Steel, Rubber-lined Steel, or Teflon-lined Steel materials of construction shall NOT be used.

Note: Approved Manufacturers – Verantis, Hartzell, Winfan, Kyowa, or approved equivalent

	<u>Solvent Exhaust Fans</u> ○ Fan housing shall be hot dipped galvanized Steel materials of construction. ○ Fan impeller shall be made of hot dipped galvanized Steel. ○ Fan be of AMCA Class A Spark Resistant Construction for all parts in contact with airstream (fan impeller, shaft, etc.) in accordance with ANSI/AMCA Standard 99-16 and AMCA Standard 99-0401-86 Note: Approved Manufacturers – New York Blower (NYB), Twin City Fan (TCF), or approved equivalent <u>General Exhaust Fans, GHVAC Exhaust Fans</u> ○ Fan housing shall be hot dipped galvanized Steel materials of construction. ○ Fan impeller shall be AMCA Class IV rated. ○ Fan impeller shall be made of hot dipped galvanized Steel. ○ Fans serving Flammable Rooms or rooms with flammable chemicals/gases shall be of AMCA Class A Spark Resistant Construction. For other fans, it is optional for fan be of AMCA Spark Resistant Construction. Note: Approved Manufacturers – New York Blower (NYB), Twin City Fan (TCF), or approved equivalent
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Table 9: MAU Equipment Selection Criteria

Equipment	Type and Description
<u>Housing</u>	• Panels for floor, walls and ceiling with a double shell with injected Polyurethane foam; fire resistance of the insulation is B1 according to DIN 4102 part1 • Both shells made of powder-coated sheet metal • Surface inside of chamber for filter 1 & 2, pre-heater, pre-cooler, washer, cooler 2 and re-heater to be made of stainless steel (SUS 304) • Panels must be thermal bridge free and self-supporting • Doors must be built like the panels and fitted with at least 2 hinges and lock bolts with exterior levers, sealing consists of a 2-lip-seal

Equipment	Type and Description
	around the doors; doors have to be built with insulation to avoid any condensation on the outside of the unit • All doors to be equipped with a barrel-lock to be lockable with one single key for all doors and units • Following unit sections/chambers shall be equipped with doors with visible window and working light with switch outside the chamber: 1. Pre-Filter 2. Secondary Filter 3. Pre-Heating Coil & Humidifier (if required, including with Freezestat) 4. Primary Cooling Coil 5. Re-Heat Coil / Reheater (if required for humidity control of the room/zone) 6. Fan • Housing to be installed on a base frame made of single galvanized steel H-beams which has to fit the different loads of the single chambers of the unit; the base frame to be assembled on site with bolts and nuts to allow the exact alignment of the frame • For the vibrations elimination will be installed one dedicated layer of pads made of vibration absorbent between the unit housing base frame and the concrete foundation; for the design of the layer of pads a calculation for the critical unit components has to be provided and approved before installation; • The static pressure test hole should be reserved before and after each component to facilitate the measurement.
<u>Heating / Cooling Coil</u>	• Frame consisting of stainless sheet steel SUS 304 • Tubes shall be 5/8-inch OD, seamless continuous copper, without flow turbulating devices, minimum 0.025-inch wall thickness, serpentine type, with return bends, same-end connections, rolled into headers with copper bushings or brazed into nonferrous headers, arranged for counter flow between air and water.

Equipment	Type and Description
	• Fins shall be minimum 0.0075-inch-thick plate-type aluminum, mechanically bonded to fins and headers. Provide corrugated plate fins to pipes. 1. Provide adequate control of fin spacing and provide a permanent mechanical bond between the pipes and the plate fins, by expanding pipes into fin collars, so that fin collars completely cover the pipes. • Casings shall be 16-gauge galvanized steel with galvanized steel center tube supports for coils over 42 inches, air bypass seals, and flanged duct connections. • Headers: 1. Headers shall be manufactured from seamless copper tube. All core joints and brass adapter connections shall be brazed with approved copper brazing alloys. 2. Provide drain plugs to bottom of headers, arranged to fully drain coils. 3. Provide inlet connections at the bottom of supply headers at the "air-off" or "air-on" face of coils (as specified) and outlet connections at the top of return headers at the "air-on" face of coils. • Fins, tubes and header shall be coated with Heresite (wherever specified), epoxy or equivalent to approval of purchaser. • Fin spacing: minimum 2 mm up to 10 mm according to detailed specification. • Air Leakage: 1. Minimize air leakage through gaps between the fin ends and frames. 2. Use baffle plates to close the space between the coil frame and the surrounding structure or equipment. 3. Seal gaps between coils and surrounding structure or equipment using non-hardening mastic. • Piping including drainage and bleeding connection with seal plug. • Operation pressure for coils 8 bar • Test pressure minimum 16 bar overpressure, leak test to be done under water according to Pressure Equipment Directive 97/23/EC and EN 378-2000. • Drain pan

Equipment	Type and Description
	1. Provide drainage from cooling coils and other components where water may collect. 2. The design should be to collect condensate and allow complete drainage so that no residue of water remains inside the unit when turned off; extend under the coil both upstream and downstream of the coil; extend under droplet separator; and extend under uninsulated piping fittings. The drain trays shall not be flat to avoid residue of water. 3. Material: should be made of Stainless steel at least SUS 304, > 3mm, in-house welded, with drainage openings. 4. Stacked coils: Provide each coil with a tray and run a minimum DN 40 drain pipe from the upper tray to the lowest tray. Acceptable Manufacturers – Aerofin, Johnson Controls, Huntair, Heatcraft, Trane, Precision Coils, or approved equal
<u>Droplet Separator</u>	• 1st Cooler will be equipped with droplet separator consisting of reinforced Polypropylene (PPTV) which allows a continuous operational temperature up to 50 ° C • Complete separation of drops at a drop size limit of 15 micron at 2.5 m/s face velocity • Separation profiles easy to be removed for cleaning and maintenance purposes
<u>Fan</u>	• Fans shall be Centrifugal type, Direct-Drive type. • Fan base frame to be equipped with spring isolators to reduce vibrations • Fan sizing and selection shall be as follows: 1. Fans shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower). 2. Fan curve shall show the performance indicated at the maximum operating point within the fan motor size limitations (flow and pressure, based on fan impeller size, VSD speed, and full brake-horsepower to

Equipment	Type and Description
	match motor size rating). Resultant fan tip speed shall be stated. 3. The maximum allowable fan impeller tip speed rating shall be published. 4. The fan maximum operating point tip speed shall be at least 15% - 20% less than the maximum allowable tip speed of the fan. Meaning, for continuous operation at the fan maximum operating point, the fan impeller tip speed shall be less than 80% - 85% of the maximum allowable fan impeller tip speed rating. 5. Fan impeller shall be AMCA rated, Class IV 6. Heavy duty bearings 7. Shaft shall have critical speed 125% of fans maximum operating speed 8. Mechanical efficiency shall be 75% above. 9. Fan assembly shall be statically and dynamically balanced. 10. Totally enclosed fan-cooled motor with over-temperature protection. 11. Sound level performance to be < 75 dBA at distance of 1m away from fan enclosure. • Fan motor sizing and selection shall be as follows: 1. Fan motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions. 2. Fan motors shall be rated for inverter duty with variable speed operation. 3. Fan motor efficiency rating shall be minimum IE3. 4. Fan motor IP rating shall be minimum IP55. Acceptable Manufacturers – New York Blower (NYB), Twin City Fan (TCF), or approved equivalent
<u>Filter</u>	• Filter frames shall be stainless steel with spring fasteners out of stainless steel. The inside flanges shall be gasketed for positive seal. • Pre-filter: rated at 25 to 35% (MERV 8 or G4)

Equipment	Type and Description
	Recommend pressure drop: 65/130 Pa (clean/dirty) Secondary filter: rated at 85% (MERV 13 or F8) Recommend pressure drop: 150/250 Pa (clean/dirty)
<u>Outlet Motorized Shut Off Damper</u>	- Frame consisting of rolled galvanized sheet steel C-profiles with slots in all corners to facilitate assembly with adjoining parts; frame assembled without any seam welding - Blades made of extruded aluminum sheath as hollow profiles with slim and streamline shaped section to give maximum rigidity against distortion and bending - Bearings for the blades made maintenance-free, consisting of oil impregnated sintered bronze bearings - Rod-linkage with minimum tolerances for installation of external actuator, including mounting base plate for actuator; no operating elements in the air stream

8.0 Duct Design and Sizing Requirements:

8.1 **Ductwork Diversity and Sizing Factor** – The Exhaust system ductwork shall be sized according to the same sizing factors stated in the above section 7.2.4 and below.

8.2 **Ductwork Mains Sizing Calculation** – Exhaust Mains shall be sized to accommodate 115-120% of the Exhaust demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM, plus the Facilities Basebuild Requirements. Therefore, the Exhaust Mains shall have 15-20% spare future capacity over the Exhaust demand at full build after initial construction of the system). See **Table 10**.

Because the Exhaust system equipment shall be installed in parallel configuration, the exhaust load shall be split evenly across the Exhaust Main ductwork, and therefore the Main ductwork shall be sized accordingly.

Table 10: Exhaust Duct Mains Sizing Calculation

- Exhaust Main Demand at Full Build, D:
 $D = (\text{TUM Peak Demand} \times 100\% \text{ Diversity}) + \text{Facilities Basebuild Requirements}$
 - Exhaust System N capacity:
 $N = 115\text{-}120\% \times D$
- Exhaust load to be split evenly across Main Ductwork

- 8.3 **Ductwork Laterals/Submains Sizing Calculation** - Each Exhaust lateral/submain serving the rooms shall be sized so that the lateral's capacity meets 115-120% of the Exhaust demand at full build for the lateral (after initial construction of the system), based on the room and equipment layout. Therefore, each Exhaust lateral/submain shall have 15-20% spare future capacity over the Exhaust demand at full build for the lateral/submain (after initial construction of the system). See **Table 11** and **Table 12** for duct sizing calculation.

Table 11: Exhaust Duct Laterals/Submain Sizing Calculation

- Exhaust Lateral/Submain Demand at Full Build (Room and Equipment),
 $DL = \text{TUM Peak Demand for Lateral} \times 100\% \text{ Diversity}$
- Exhaust Lateral/Submain Duct Sizing = $115\text{-}120\% \times DL$

Table 12: MAU Duct Laterals Sizing Calculation

- MUA Mains Demand at Full Build, D:
 $D = \text{Total Normal Exhaust} + \text{Total Emergency Exhaust}$
 1. Total Normal Exhaust = Total Exhaust in Normal mode inclusive Central Exhaust System / Dedicated Exhaust System for ALL the rooms being served by the MAU.
 2. Total Emergency Exhaust = Total Exhaust in Emergency mode inclusive Central Exhaust System / Dedicated Exhaust System for the LARGEST room / compartment being served by the MAU
 - MUA System N capacity:
 $N = 115\text{-}120\% \times D$
 - MUA Main Duct Sizing to include future units:
Sizing = N
- MAU load to be split evenly across MUA Ductwork

- 8.4 **Duct Sizing Criteria** – Exhaust Main Duct Sizing shall be selected based on the full build capacity of the Exhaust system equipment. It is recommended for the duct sizing to be designed with maximum velocities not to exceed the tables below: See **Table 13**.

Table 13 – Duct Sizing Criteria for Exhaust Ductwork

Duct Size	Maximum Velocity
10" to 18" (DN250 to DN450)	Less than maximum of 1,771 fpm (9.0 m/s)
20" to 26" (DN500 to DN 650)	Less than maximum of 1,968 fpm (10.0 m/s)
28" and larger (DN700 and larger)	Less than maximum of 2,363 fpm (12.0 m/s)

- 8.4.1 If the ductwork sizing is designed with velocities exceeding **Table 13**, the ductwork system pressure drop calculations shall demonstrate that the exhaust fan selections have adequate performance rating to meet the design parameters and system requirements.
- 8.5 **Duct Pressure Rating**, – Central Exhaust ductwork mains shall be designed to meet the minimum pressure class of -15 inches w.g. (-3,750 Pa), and submains/laterals shall be designed to meet the minimum pressure class of -8 inches w.g. (-2,000 Pa). If the system pressure at the main ductwork exceeds this amount, then the ductwork pressure rating shall meet or exceed the system pressure.
- 8.5.1 MUA ductwork mains shall be designed to meet the minimum pressure class of +4 inches w.g. (+1,000 Pa).
Dedicated Exhaust System ductwork mains shall be designed to meet the minimum pressure class of -4 inches w.g. (-1,000 Pa).
- 8.6 **Duct Supply / Exhaust Air Grille** – Supply and Exhaust Air Grille should evenly be distributed to have an effective air distribution within the room.
- 8.6.1 Exhaust Air Grille should be positioned within the top part of the room (within 300mm from the level of soffit / ceiling) if the HPM Chemical / Gas is lighter-than-air when it is leaking and mixing with the room air.
- 8.6.2 Exhaust Air Grille should be positioned within the bottom part of the room (within 300mm from finished floor level) if the HPM Chemical / Gas is heavier-than-air when it is leaking and mixing with the room air.

- 8.6.3 Supply Air Grille should be strategically positioned to avoid being exhausted before it is effectively mixed with the room air (i.e air path short-circuit).

8.7 Materials of Construction

- 8.7.1 **Acid / Ammonia Exhaust Ductwork** – Ductwork shall be FM Approved Fluoropolymer-coated Stainless Steel, FM Approved Fiberglass Reinforced Plastic (FRP), or standard Fiberglass reinforced Plastic (requiring internal sprinklers) materials of construction.

- Fluoropolymer-coated Stainless Steel ductwork includes ETFE (Ethylene-Tetrafluoroethylene) and ECTFE (Ethylene-Chlorotrifluoroethylene) coating materials.
- It is recommended to use FM Approved ductwork materials, in lieu of standard FRP ductwork that is not approved by FM. Ductwork that is FM Approved per FM Standards 4922 and 4910 do not require internal sprinklers, per NFPA Standard 318 and FM requirements.

Note: Ductwork made of PVC, PP or PE shall NOT be allowed for sizes 6" and larger. The exception is smaller size PVC ductwork, sizes 4" (DN100) and below, which is typically used for tool hookup ductwork and VMB ductwork.

- **Gaskets, general:** In general, PTFE (Polytetrafluoroethylene, aka Teflon) tape gaskets or rope gaskets shall be used for Acid and Ammonia Exhaust. Gasket shall be installed on the duct flange lip surface.

- 8.7.2 **General Exhaust Ductwork** – Ductwork shall be Galvanized Steel.

- **Gaskets:** In general, EPDM gaskets shall be used for General Exhaust.

- 8.7.3 **Solvent Exhaust Ductwork** – Ductwork shall be Stainless Steel or Teflon-coated Stainless Steel.

- **Gaskets, general:** In general, Teflon (PTFE) gaskets shall be used for Solvent Exhaust. Gasket shall be installed on the duct flange lip surface.

- 8.7.4 **MUA / GHVAC Exhaust Ductwork**– Ductwork, rectangular and circular, shall be made of Galvanized Steel. MUA ductwork to be insulated from the equipment unit outlet connection up to the supply air grille in the rooms. The ductwork insulation material

shall be vapor-tight, with a closed cell structure, made of expanded cellular polyethylene foam.

- **Gaskets:** Neoprene tape, silicon-free gaskets shall be used for MUA ductwork. EPDM gaskets shall be used for GHVAC Exhaust.

- 8.7.5 **Exhaust Ductwork Drain Piping** – piping shall be Schedule 80 PVC materials of construction. Clear PVC Schedule 40 to be used as sight glass at drain trap discharge valve for visual inspection.

8.8 Ductwork Installation

- 8.8.1 For exhaust ductwork, it is recommended to use round ductwork where possible. If this is not possible due to space constraints, rectangular ductwork is acceptable.
- 8.8.2 Ductwork branch lines and take-offs shall use increased area openings, such as conical or bell-mouth fittings (bull head tees shall not be allowed).
- 8.8.3 For exhaust ductwork, ductwork takeoffs for branch ducts, spare dampers, POCs, etc. shall face directly sideways (with higher bottom surface elevation than the main) or upwards. Ductwork takeoffs shall not face downwards or diagonally downwards. The bottom surface elevation of the takeoffs shall not be at the same height or lower than the bottom surface elevation of the main. This is to prevent any liquid from entering or accumulating in the spare damper or lateral/submain from the main.
- 8.8.4 Inlet ductwork to Exhaust Scrubbers shall be designed so that any liquid or condensation in the exhaust ductwork will drain back into scrubber sump. The scrubber inlet elevation shall be at a lower height than the exhaust duct header elevation, to prevent any liquid from entering or accumulating in the exhaust duct header.
- 8.8.5 FRP ductwork located outdoors shall be painted for UV protection.
- 8.8.6 Ductwork shall be installed in accordance with SMACNA "Industrial Duct Construction Standards."
- 8.8.7 Prior to delivery to site from the factory, 100% of the Coated Stainless-Steel ductwork shall be factory spark tested to identify any flaws, defects, or damage of the duct coating.
- 8.8.8 Immediately prior to installation on-site, a minimum of 25% of the Coated Stainless-Steel ductwork shall be spark tested on-site to identify any flaws, defects, or damage of the duct coating.

8.9 Provisions for Main Expansion

- 8.9.1 At locations in which there is a planned future extension of the exhaust main ducts, headers, or laterals/sub-laterals, provide full-size blast gates with capped blind flanges at the ductwork ends, to allow easier extension without interruption to the system operation.
- 8.9.2 At locations of unplanned future extension, provide full-size blind flanges at the ductwork ends.

8.10 Spare Dampers for Future Laterals/Submains

- 8.10.1 Additional spare dampers shall be included on the main ductwork, to allow installation of future laterals/submains for capacity expansion. Based on the number of laterals/submains included in the design, a minimum quantity of 10-15% of additional spare dampers for future laterals shall be included.
- 8.10.2 Only Spare Dampers facing directly sideways (with higher bottom surface elevation than the main) or upwards shall be used. Spare Dampers shall not face downwards or diagonally downwards. The bottom surface elevation of the spare dampers shall not be at the same height or lower than the bottom surface elevation of the main. This is to prevent any liquid from entering or accumulating in the spare damper or lateral/submain from the main.

8.11 Lateral Point-of-Connection Quantity and Sizes

- 8.11.1 The laterals shall have sufficient quantity of Point of Connections (POCs) to meet the tool / cabinet layout and requirements.
- 8.11.2 Provide adequate spacing in between each POC, to allow proper airflow pattern and uniformity.
- 8.11.3 The sizes for each POC shall be checked and verified versus the actual tool requirements. Minimum POC size shall be DN150 (6"), but larger POC sizes can be provided, based on the actual tool / cabinet requirements.
- 8.11.4 Only POCs facing directly sideways (with higher bottom surface elevation than the lateral/submain) or facing upwards shall be used. POCs shall not face downwards or diagonally downwards. The bottom surface elevation of the POC takeoffs shall not be at the same height or lower than the bottom surface elevation of the lateral/submain. This is to prevent any liquid from entering or accumulating in the POC and tool hookup ductwork from the lateral/submain.

- 8.12 **Exhaust Duct Drains** – Refer to the Global Facilities -Mechanical – Process Exhaust System Standard for requirements of Duct Drains on Process Exhaust ductworks. Duct drains shall be discharged to the dedicated compatible chemical waste drain point.
- 8.13 **Exhaust Fire Protection Sprinklers** – Refer to the Global Facilities – Mechanical – Process Exhaust System Standard for requirements of Fire Protection Sprinklers on Process Exhaust ductwork.
- 8.14 **Ductwork Test Ports, Inspection Ports and Sampling Ports** Ductwork test ports shall be provided to allow for airflow measurement, pressure reading and air sampling.
- 8.14.1 Main Ductwork – Each main shall have test ports at both ends of the main to allow airflow and pressure readings.
- 8.14.2 Lateral/Submain Ductwork – Each lateral/submain shall have test ports at both ends of the lateral/submain to allow airflow and pressure readings.
- 8.14.3 Exhaust Fans - Each exhaust fan shall have test ports upstream and downstream of the fan to allow airflow and pressure readings.
- 8.14.4 Test ports shall be minimum 25mm (1 inch) diameter in size.
- 8.15 **Fire Dampers**
- 8.15.1 Fire Dampers shall not be installed in Exhaust ducts, unless required by local code or jurisdiction.
- Fire dampers inside Exhaust ducts are prohibited by NFPA Standard 318 Section 5.3.5, and International Mechanical Code Section 510.6.1
- Note:** If the local code or jurisdiction requires fire dampers in the exhaust ductwork, it shall take precedence and the fire dampers shall be installed.
- 8.16 **Other Accessories** - Provide other accessories as outlined in **Table 11**.

Table 11: Accessories to be installed

Accessories	Type & Installation Location
Ductwork Insulation	Ductworks shall be insulated wherever there is potential occurrence of air condensation (internal or external of ductworks)

Accessories	Type & Installation Location
Ductwork Supports	• Provide duct hanger and support material gauges in compliance with details on the Drawings. Where details are not provided, comply with SMACNA Industrial Duct Construction Standards. • Ductwork hanger and suspension material may be threaded rod, structural shapes, or strut material. The use of wire as suspension material is prohibited. • Ductwork hanger supports and spacing shall take into account the potential weight of water when the ductwork contains fire sprinklers. • Ductwork supports shall take into consideration of potential corrosive room environment. Material selection and/or special treatment of the support material to be provided. • Refer to the specific project's Construction Specification, (i.e. Spec Section 15140 or equivalent Spec) for additional requirements.
Labeling and Identification	• Provide identification labels at minimum of every 6m distance and change of direction • Refer to the specific project's Construction Specification, (i.e. Spec Section 15190 or equivalent Spec) for additional requirements. <u>Ductwork</u> • Labels on Exhaust Ductwork shall be Yellow Background with Black Lettering • "AE" nomenclature shall be used for Acid Exhaust • "AME" nomenclature shall be used for Ammonia Exhaust (also Alkali Exhaust) • "SE" nomenclature shall be used for Solvent Exhaust (also known as VOC Exhaust) • "GE" nomenclature shall be used for General Exhaust • "GHVAC Exhaust" nomenclature shall be used for General HVAC Exhaust • "MAU" nomenclature shall be used for Makeup Air Ductwork <u>Piping</u> • Labels on Exhaust Drain Piping shall be Yellow Background with Black Lettering

9.0 Sustainability Design:

- 9.1 **Exhaust Fan Parallel Header Design** – Exhaust Fans shall be arranged in parallel and connected to a common ductwork header upstream of the fans and downstream of the Scrubbers/Abatement Units. This allows multiple exhaust fans to operate together in parallel, to lower the fan operating speed and fan energy consumption.
- 9.2 **Ductwork Pressure Drop Sizing** – It is recommended to increase the ductwork sizes where possible, to minimize the air velocity and pressure drop through the ductwork. This will result in lower exhaust fan energy consumption.

10.0 Electrical Design:

- 10.1 **Electrical power supply system for HVAC System, Exhaust System and all other equipment shall be designed and installed in compliance with **Global Facilities - Design & Construction Standard - Electrical - Electrical Design Specification**.**
- 10.2 **Power Segregation** – Electrical power supply for exhaust fans shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.
- 10.3 **Un-interrupted Power for Controls** – 100% of exhaust system controls including PLCs, LCPs, and other auxiliary control power shall be connected to an Un-interrupted Power source to allow continuous operation of the exhaust system during momentary power interruption or voltage sags, spikes, and fluctuation events.
- 10.4 **Emergency Power** – 100% of the exhaust fans of the Emergency (Dedicated) Exhaust System shall be connected to Emergency Power.

A minimum of 50% of the Normal Exhaust fans (via the Central Process Exhaust System) of each system shall be connected to Emergency Power.

This is per requirements of Factory Mutual (FM) Standard 1056, National Fire Protection Agency (NFPA) Standard 318, and International Building Code (IBC) Section 415, to maintain 50% capacity (exhaust airflow) upon a power loss.

The interpretation is that by having 50% of the exhaust fans on emergency power, it is sufficient to provide approximately ~50% capacity (exhaust airflow) upon a power loss, to satisfy the requirements.

To maintain proper pressurization levels, 100% of MAU units (dedicated) that supplies Makeup Air to HPM Chemical / Gas Room shall also be on emergency power.

10.5 Variable Frequency Drives – All fans (Central or Dedicated System) shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual fan motor size.

10.5.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

10.5.2 Variable Frequency Drives shall be configured to “Fail Last Speed” configuration upon any failure, including but not limited to, loss of power, loss of communications, etc.

10.6 Motors

10.6.1 Exhaust fan motors shall be inverter duty rated with insulated bearings and shaft grounding kits. Motors shall be compatible with the associated VFD.

10.6.2 All motors located outdoors shall be IP56 rated protection.

10.6.3 All motors shall be National Electrical Manufacturers Association (NEMA) IE3 rated or better.

10.6.4 All exhaust fan motors that serve HPM Chemical / Gas Rooms, where there is risk of flammable gas leakage, must be “spark-proof” and suitable to be used in Class 1 Division 2 hazardous area.

10.7 Local Disconnect Switches – Each exhaust fan shall have a lockable type, local disconnect switch installed adjacent to the fan, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the fan.

10.7.1 All disconnect switches located outdoors shall be IP56 rated protection.

10.7.2 All disconnect switches must be reasonably accessible for the authorized personnel to safely isolate the power supply to facilitate maintenance works.

10.7.3 All local disconnect switches located in HPM Chemical / Gas Room where there is risk of flammable gas leakage must be “spark-proof” and suitable to be used in Class 1 Division 2 hazardous area.

11.0 Instrumentation:

11.1 Instrumentation for HVAC System, Exhaust System and all other equipment shall be designed and installed in compliance with Global Facilities - Design & Construction Standard - I&C – Instrumentation.

11.2 The following Instrumentation shall be installed and monitored/controlled via FMCS: See **Table 12** and **Table 13**.

Table 12 – Instrumentation Devices

Accessories	Type & Installation Location
Pressure Transmitters (central process exhaust system)	Refer to Global Facilities – Mechanical – Process Exhaust System Standard for Pressure Transmitter requirements on Central Process Exhaust System
Pressure Transmitters (General HVAC exhaust system)	General HVAC Exhaust: Exhaust Duct Main header upstream of the Exhaust Fans shall have a minimum of one pressure sensing locations at a distance of 3/4 of full ductwork length to the end of line. Each pressure sensing location shall have two pressure transmitters, one duty and one redundant backup (for N+1).
Exhaust Flowmeters	Install a flowmeter on each exhaust fan to allow continuous exhaust load monitoring.
Additional Spare Ports	All exhaust laterals shall have a spare port and installed at the lateral end-of-line, to allow for future pressure transmitters to be installed. The future pressure transmitters will allow continuous SCADA monitoring of each lateral pressure.

Table 13 - Instrumentation for MAU

Instrumentation	Type and Installation Location
Pressure Transmitters	Install at Main header at a distance of 3/4 of full ductwork length to the end of line. (Quantity: 1 duty and 1 backup [2 Total])
Differential Pressure Transmitters	Install at each filter bank (Quantity: 1 duty)
MAU Discharge Temperature/Humidity Transmitters	Install at outlet of MAU (Quantity: 1 duty) Can be convert to Dewpoint & Enthalpy
Temperature Transmitter	Install at outlet of heater and Cooler (Quantity: 1 duty)
Room HVAC (Temperature /Humidity Transmitter)	HPM Chemical / Gas Room shall have air temperature and humidity sensors for SCADA monitoring and control purposes

- 11.3 Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.
- 11.4 Gas detector / instrument that is monitoring for gas leakage or fire shall be located as such that the detection signal capable of identifying the specific zone of the incident.
- 11.5 The Pressure Differential Transmitters on exhaust ductwork shall reference the Zero Reference Pipe (ZRP). This shall be the same ZRP used for Make-up air unit pressurization reference.
- 11.5.1 ZRP shall reference outdoor atmospheric pressure. ZRP external probe shall be located in a place with minimum wind interference and shall have adequate cover protection from wind.

12.0 Control Sequence Philosophy:

12.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 14**.

Table 14: System Control Logic

Control Parameter	Control Logic
Central (Process) Exhaust System Pressure Control	- Speed output of Exhaust Fan VSDs shall be modulated to maintain the pressure setpoint at the Exhaust Main duct. - The pressure setpoint shall be averaged across the number of pressure sensing locations spread across the Exhaust Main duct.
Central (Process) Exhaust Fans Start/Stop	Exhaust Fan Start/Stop shall be done automatically via SCADA based on required number of running fans
Exhaust Fan Start / Stop (Emergency Dedicated system)	- Exhaust Fan Start/Stop shall be done automatically via SCADA or Hardwired Logic, during emergency mode. - Exhaust Fan Start/Stop/Rev shall be capable of being remotely controlled via SCADA by Operation Personnel. Emergency mode exhaust shall be activated in the following ways: - Automatically in a fire event or Chemical / Gas leak event. - Remotely by Operation through SCADA. - Locally through a manual emergency
MAU	- MAU Start/Stop shall be done automatically via SCADA based on required number of running units VSD Speed of MAU Fans shall be modulated to maintain the pressure setpoint at the Main duct. Preheating (if required) - Preheating coil (if necessary. Coil before the 1st cooling coil) temperature control valve shall be

Control Parameter	Control Logic
	modulated to pre-heat the outdoor air to the temperature set point at outlet of preheating coil. Humidifier (if required) - Humidifier (if necessary. Install after the Preheating coil) control valve shall be modulated to the humidity set point at the outlet of the humidifier. Cooling / Dehumidification - Position output of 1st cooling coil temperature control valve shall be modulated to maintain the temperature (and dewpoint) setpoint at outlet of 1st cooling coil. Reheating (if required) - Position output of 2nd coil (i.e. Re-heater) control valves shall be modulated to maintain the relative humidity (RH%) setpoint at outlet of MAU.
Ductwork Motorized Dampers	- Motorized dampers opening percentage to be measured and confirmed through an air-balancing activity. - Motorized dampers opening percentage to be actuated to rated percentage, either during Normal or Emergency mode, to allowed for proper air flowrate. - Normal / Emergency mode rated flowrate to be validated by manual measurement.

12.2 A detailed Sequence of Operation (SOO) shall be developed specifically for each system, as part of each construction project.

12.3 Control Valves and Dampers

12.3.1 The following control valve and damper fail positions shall be configured. See **Table 15**.

Table 15: Emergency Exhaust System Motorized Dampers and Control Valves

Control Valve or Damper	Control Valve or Damper Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
Exhaust / MAU Ductworks Motorized Dampers	Fail Last State	Fail Last State	Fail Last State
Control Damper at Exhaust Fan inlet	Fail Last State	Fail Last State	Fail Last State
MAU Temperature Control Valves	Fail Last State	Fail Last State	Fail Last State

12.3.2 The control valves and dampers shall be double-acting type actuators, capable of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

12.3.3 After a loss event of control air, power, or PLC signal, and after it is restored, the control valve or damper output command/position shall be set to the same command/position it was operating at prior to the loss event.

12.3.4 All control valves and dampers shall have a manual override bypass, to allow manual operation of the valve or damper upon a failure.

12.3.5 For motorized dampers with actuators, it is recommended to have square-shaped shafts connecting the damper and the actuator, to prevent slippage and mis-operation. Prior to system startup, perform inspection and operations check of all dampers and actuators including shaft, linkages, and connection points.

13.0 Inter-Discipline Coordination Matrix:

13.1 The following provisions shall be provided by each of the following disciplines, and coordinated accordingly: See **Table 16**.

Table 16: Inter-Discipline Coordination Matrix

Discipline	Provision
Architectural	- HPM Chemical / Gas room shall be constructed with appropriate fire compartmentation and fire rated partition / wall. - Cut openings in walls and floors for ducting penetrations, and seal openings after duct installation - Provide adequate fire stopping at all required fire wall penetrations
Mechanical	- Louver shall be constructed to blend in Architectural façade design, e.g. color, structure, etc. - Exhaust louver shall be constructed at least 5 meter away from any fresh air intake / building opening. - Fresh air louver shall be constructed at least 5 meter away from any exhaust louver.
Electrical	- Provide local disconnect switches for all exhaust fan motors and pump motors. - Provide power wiring for all Electrical equipment/ components - Provide grounding for all Electrical equipment/ components - Provide Emergency Power / UPS Power as outlined in this Standard - Provide Power Segregation as outlined in this Standard - Provide VFDs for all exhaust fan motors. - Provide lightning protection for outdoor equipment including exhaust discharge stacks.
Controls	- Provide control damper and control valve actuators and positioners - Provide control wiring - Provide instrument control air tubing - Provide Zero Reference Pipe (ZRP) for atmospheric pressure reference of Exhaust and Makeup Air Systems - Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, gas detector, fire detector etc. - Provide FMCS/SCADA system connectivity of all Local Control Panels

14.0 Quality Inspection and Testing:

14.1 Inspection and Test Plan (ITP) – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

14.1.1 Ductwork

- Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Assembly Procedures, Qualifications, and Inspection
- Duct Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Ductwork Coating Spark Testing - Immediately prior to installation on-site, a minimum of 25% of the Coated Stainless-Steel ductwork shall be spark tested on-site to identify any flaws, defects, or damage of the duct coating.
- Ductwork Leak Testing –100% of HPM Chemical / Gas Room HVAC ductwork shall be leak tested.
- Leak Testing and Pressure Testing
- Leak testing of condensate drain pipework including its pipe joint with cooling coil, ductwork, and etc
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

14.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test (FAT)
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.

14.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist

- Training, Documentation, and Handover

15.0 Startup Commissioning / Site Acceptance Test:

15.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Testing & Commissioning / Acceptance Tests.

15.1.1 Fans

- Fan Motor Rotation Check
- Fan/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Fan Standalone Startup Procedure per Manufacturer's procedure
- Fan Vibration Level Checks
- Verify Fans Operation through its operating range (0 to 50/60 Hz)
- Verify Fan Start/Stop Function to Rotate Operating Fan
- Verify Fan Auto-Start sequencing, upon failure of an operating Fan
- Verify Fan Speed control in response to Exhaust pressure fluctuations
- Verify Fan Inlet Motorized Damper opens upon fan startup and closes upon fan stop, while keeping exhaust pressure stable
- Confirm Exhaust system pressure control is stable during fan manual start/stop rotation
- Confirm Exhaust system pressure control is stable during fan fault/trip event and next fan auto-start sequencing
- Verify VSD fails last speed upon any failure
- Verify control interface and communication checks to FMCS/SCADA
- Control interface and communication checks to Local Control Panel, PLC, and FMCS/SCADA.

15.1.2 MAU

- Confirm the external static pressure and the flowrate
- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test

15.1.3 Integrated system performance test

- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test
- Emergency mode exhaust system integrated testing, exhaust fans kick-in and MAU ramp up, control / interlock with motorized dampers activation

16.0 Reference Specifications:

- 16.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 17**.

Table 17 – Specification Sections

Spec Section	Spec Contents
Section 15060	Pipe and Pipe Fittings
Section 15190	PVC and CPVC Piping Systems
Section 15101	Valves
Section 15120	Piping Specialties
Section 15121	Process piping Specialties
Section 15130	Indicating Devices
Section 15140	Supports and Anchors
Section 15190	Mechanical Identification
Section 15240	Mechanical Sound and Vibration Control
Section 15260	Piping Insulation
Section 15280	Equipment Insulation
Section 15330	Wet Pipe Sprinkler Systems
Section 15461	Process Exhaust System
Section 15515	Hydronic Specialties
Section 15540	Pumps
Section 15860	Exhaust Fans
Section 15861	FRP Exhaust Fans and Stacks
Section 15891	Metal HVAC Ductwork and Accessories
Section 15892	Metal Industrial Ductwork and Accessories
Section 15993	Air Systems Testing, Adjusting, and Balancing
Section 15170	Motors
Section 15171	Variable Frequency Drives

17.0 Document Control

17.1 Approval: GLOBAL_FAC_GFTT_APPR

17.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

17.3 Final Viewer Access

All Micron sites Facilities Team Members

17.4 Retention

Adherence to the requirements specified in the
[Global Facilities - Document Control Procedures](#)

17.5 Review

Biennially (two years)

18.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	04/05/21
1	Section 6	6.4.3: Added Note 4 for Table 1 6.5.7: Added new sub-session	03/21/22
	Section 10	10.1: Added new sub-session	
	Section 11	11.1: Added new sub-session	
	Section 14	14.1.1: Added "Leak testing of condensate drain pipework including its pipe joint with cooling coil, ductwork, and etc"	